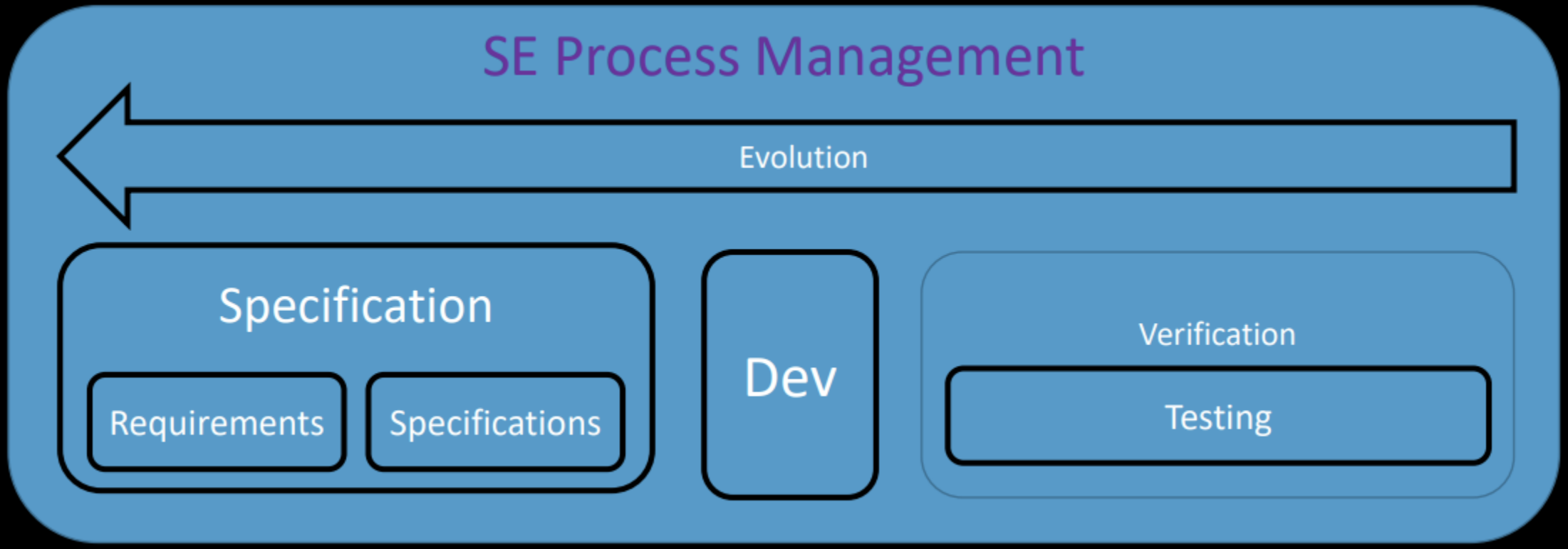
Software Engineering Management COMP4135

Seminar 8

Risk Management

Risk Management

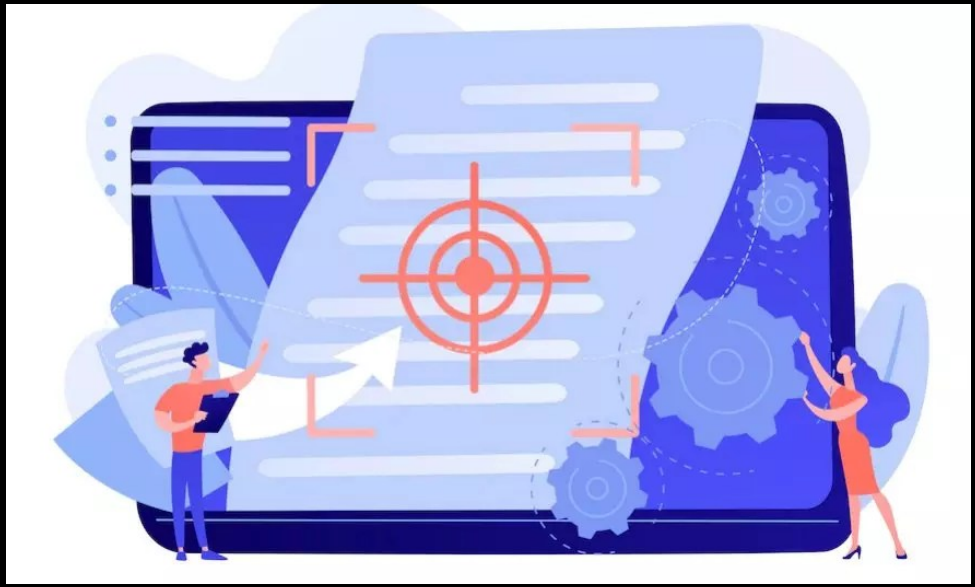


Agile Principles

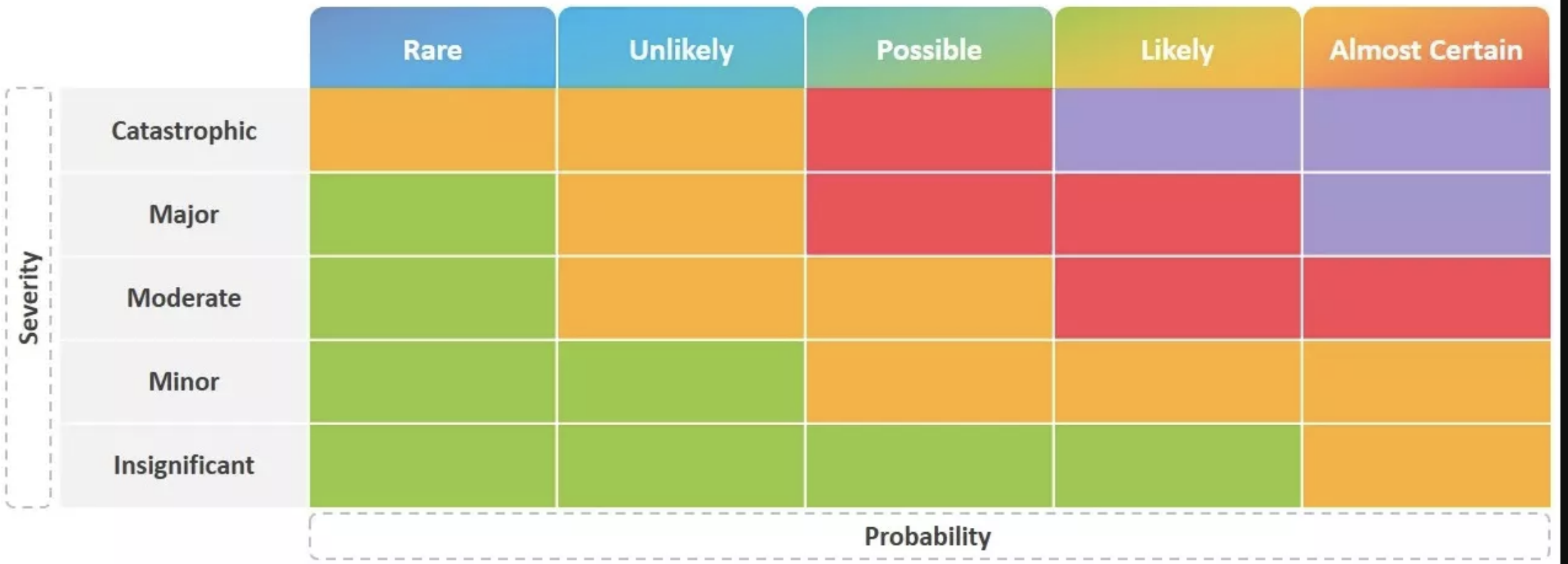
Agile is a set of values and principles

Working software over comprehensive documentation

Responding to change over following a plan



Example of Risk management plan



Principles

- Global Perspective: In this step of project management, we will go over the overall system description, design, and implementation. We consider the risk and its potential consequences.
- Keep an eye on the future: Consider the threat that might appear in the future and make plans to direct the next events.
- Open Communication: This allows for the free flow of information between the client and the team members, allowing them to be confident about the risks.
- Integrated management: Risk management is integrated into project management in this method.
- Continuous process: The risks are tracked constantly throughout the risk management paradigm during this phase.

Steps of Risk Management

1. Risk Identification
2. Risk Assessment and Prioritization
3. Risk Avoidance and Mitigation
4. Risk transfer
5. Risk acceptance
6. Risk Monitoring

The Scrum Framework At a Glance

Product Backlog

(owned by PO)

The single source of work for the team, the product backlog is an emergent, ordered list of what is needed to improve the product and includes the product goal.

Refinement

Refinement is an ongoing activity used to add details, including size and order, to items in the product backlog.

Sprint Planning

To start the sprint, sprint planning establishes the sprint goal, what can be done, and how the chosen work will be completed.

Sprint Backlog

(owned by the developers)

The sprint backlog is a real-time, emergent picture of work that shows the product backlog items selected for the sprint by the developers, plus a plan for delivering the product increment and realizing the sprint goal.

*Sprint of 1 month or less

(An inspect and adapt cycle that turns ideas into value)

Each sprint should bring the product closer to the overall product goal.

Daily Scrum

The purpose of the daily scrum is for developers to inspect progress toward the sprint goal and adapt the sprint backlog as necessary, adjusting the upcoming planned work.

Sprint Review

The purpose of the sprint review is to inspect the outcome of the sprint and progress toward the product goal with key stakeholders, while determining future adaptations.

Increment

A sum of usable sprint backlog items completed by the developers in the sprint that meets the definition of done. Each increment should work with all prior increments.

Sprint Retrospective

The sprint retrospective is for the scrum team to inspect how the last sprint went with regards to individuals, interactions, processes, tools, and definition of done. The team identifies improvements to effectiveness, addressing the most impactful improvements as soon as possible.

Product Owner | **Scrum Master** | **Developers**

Scrum artifacts that help manage the work | Events that occur inside each sprint | Ongoing activity

Who should participate and for how long?

SPRINT PLANNING	DAILY SCRUM	SPRINT REVIEW	SPRINT RETROSPECTIVE
Who: The entire scrum team. **Timebox:** Maximum of 8 hours for a month-long sprint. Shorter timebox for shorter sprints.	**Who:** The developers, but if the product owner or scrum master are actively working on items in the sprint backlog, they participate as developers. **Timebox:** 15 minutes.	**Who:** The entire scrum team. Stakeholders are invited to provide feedback on the increment. **Timebox:** Maximum of 4 hours for a month-long sprint. Shorter timebox for shorter sprints.	**Who:** The entire scrum team. **Timebox:** Maximum of 3 hours meeting for a month-long sprint. Shorter timebox for shorter sprints.

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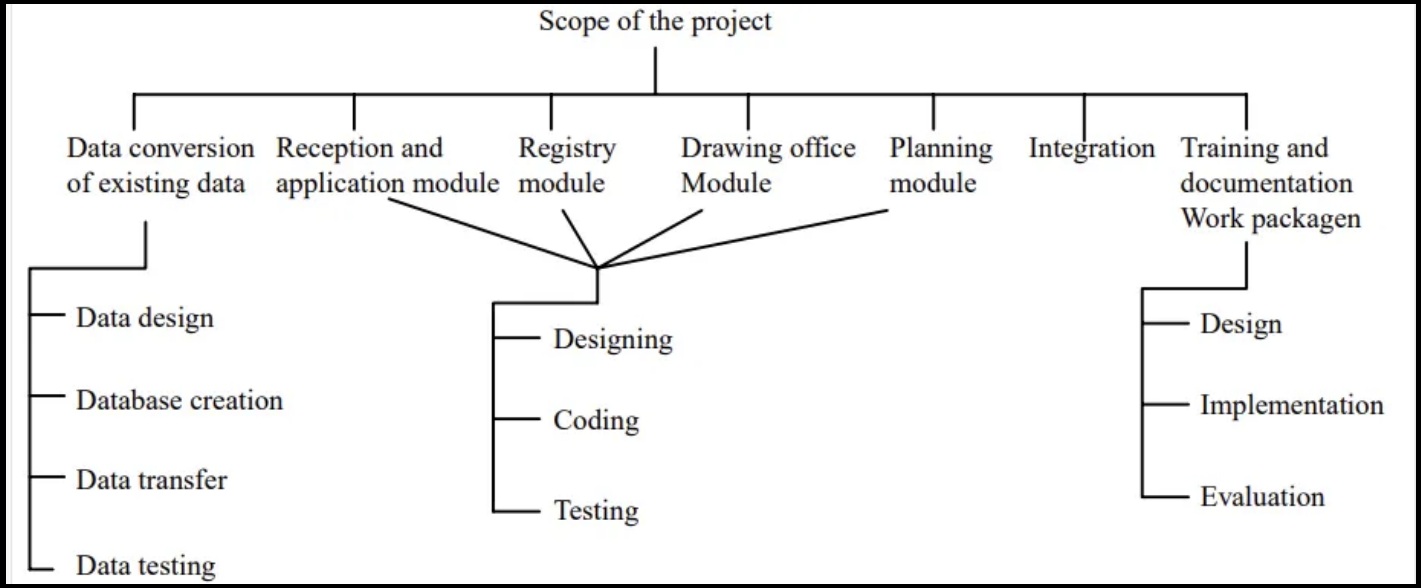
Group Discussion

The Town and Country Planning Office (TCPO) is the government agency that regulates building construction in Barbados. This department receives up to 3,000 applications a month for permission to construct buildings. Currently an application to build takes between three months to three years to reach the approval stage. This is partly due to inadequate application tracking procedures. The department planned a software project for application tracking management system. The estimated cost and planned duration of the project are $400,000 and 12 months, respectively.The contract had the following stipulation (Town and Country Planning Application Processing System Tender Document, 2001): In the case of a delay of more than one month the supplier shall be liable to pay damages calculated from the expiry of the contractual period for each day the delay lasts, such damages to be fixed at 1/1,000 of the value of the undelivered software per month.



Group Discussion

1. **Incorrect requirements/specification.** The requirements phase of a software project is one of the most crucial phases. If the requirements are not exact then the application will not meet the needs of the users.
2. **Incompatible development environment.** The language used to develop the software does not easily lend itself to the development of the particular application.
3. **Inadequate design.** The design of the database and or the data structures does not adequately cover all the data to be processed by the system.
4. **Loss/lack of resources**. The Loss or unavailability of key personnel during the course of the project.
5. **Unavailable customer contact**. Effective communication between client/customers and developer throughout the software development helps in achieving success.
6. **Scope Creep**. The requirements of the project are continually added to thus causing scope creep
7. **Problems in coding and unit test.** Quality of coding is judged by testing. Various types of testing are planned to ensure quality in software development depending on heir characteristics. However, if the programmers, who are developing the codes, are held responsible for testing, it is difficult to properly check the quality. Hence, it is always suggested to establish a different group for unit testing.

Discussion: Risk Mapping

Risk Event	Probability (Very Low/Low/Med/High/Very High)	Severity (Very Low/Low/Med/High/Very High)	SPR
Loss of key personnel			
Incorrect requirements			
Scope creep			
Inadequate design			
Unavailable customer contact			
Incompatible dev environment			
Coding/unit test problems			

Discussion: Risk Mapping

Risk Event	Mitigation Strategy	Strategy type
Loss of key personnel	Loosing resources at critical points of software development has high negative impact. This can be avoided by ensuring that project team members au fait with all aspects of the project	
Scope creep	Scope creep plagues most projects (especially software development) and causes time and cost overrun. However, scope creep is sometimes needed to address quality issues in software development. A dynamic scope management plan with involvement of client will improve project performance	
Incorrect requirements or specifications	Incorrect specifications of software projects are major issues in managing software development projects, as clearly establishing client's requirements is not always an easy task. Technique like process reengineering and benchmarking can be employed for deriving the requirements with the active involvement of functional people of the client's organizations	
Inadequate Design	This type of problem is present in any software development project. Use software modelling tools helps reduce the effect of inadequate design	
Incompatible development environment	Researching all limitations of development environment before implementation helps reducing risk drastically	
Unavailable customer contact	Client's involvement in software development is one of the most important factors of success. Effective communication between developer and client can be maintained using information technology, e.g. e-mail, web site	
Problems in coding and unit test	Developing a software inspection process and ensuring independent testing can reduce the problems related to coding and unit test	

Case Study

Severity probability factor rating (SPR)	
4	Avoid these risks all together. Recognise them from the start and plan to avoid them from occurring
3	Mitigation strategy and detailed contingency plan
2	Mitigation strategy and outline contingency plan
1	Mitigation strategy
0	Treat as a project assumption

Severity	*Very high*	3	3	3	3	4
	High	2	2	2	3	3
	Medium	1	2	2	2	3
	Low	1	1	2	2	3
	Very low	0	1	1	2	3
		Very low	*Low*	*Medium*	*High*	*Very high*
			Probability			

Risk events	Risk strategy	Strategy type
Loss/lack of resources	Loosing resources at critical points of software development has high negative impact. This can be avoided by ensuring that project team members au fait with all aspects of the project	Avoidance
Scope creep	Scope creep plagues most projects (especially software development) and causes time and cost overrun. However, scope creep is sometimes needed to address quality issues in software development. A dynamic scope management plan with involvement of client will improve project performance	Transference/reduction
Incorrect requirements or specifications	Incorrect specifications of software projects are major issues in managing software development projects, as clearly establishing client's requirements is not always an easy task. Technique like process reengineering and benchmarking can be employed for deriving the requirements with the active involvement of functional people of the client's organizations	Transference/reduction
Inadequate Design	This type of problem is present in any software development project. Use software modelling tools helps reduce the effect of inadequate design.	Reduction
Incompatible development environment	Researching all limitations of development environment before implementation helps reducing risk drastically	Reduction
Unavailable customer contact	Client's involvement in software development is one of the most important factors of success. Effective communication between developer and client can be maintained using information technology, e.g. e-mail, web site	Transference/reduction
Problems in coding and unit test	Developing a software inspection process and ensuring independent testing can reduce the problems related to coding and unit test	Reduction

Results

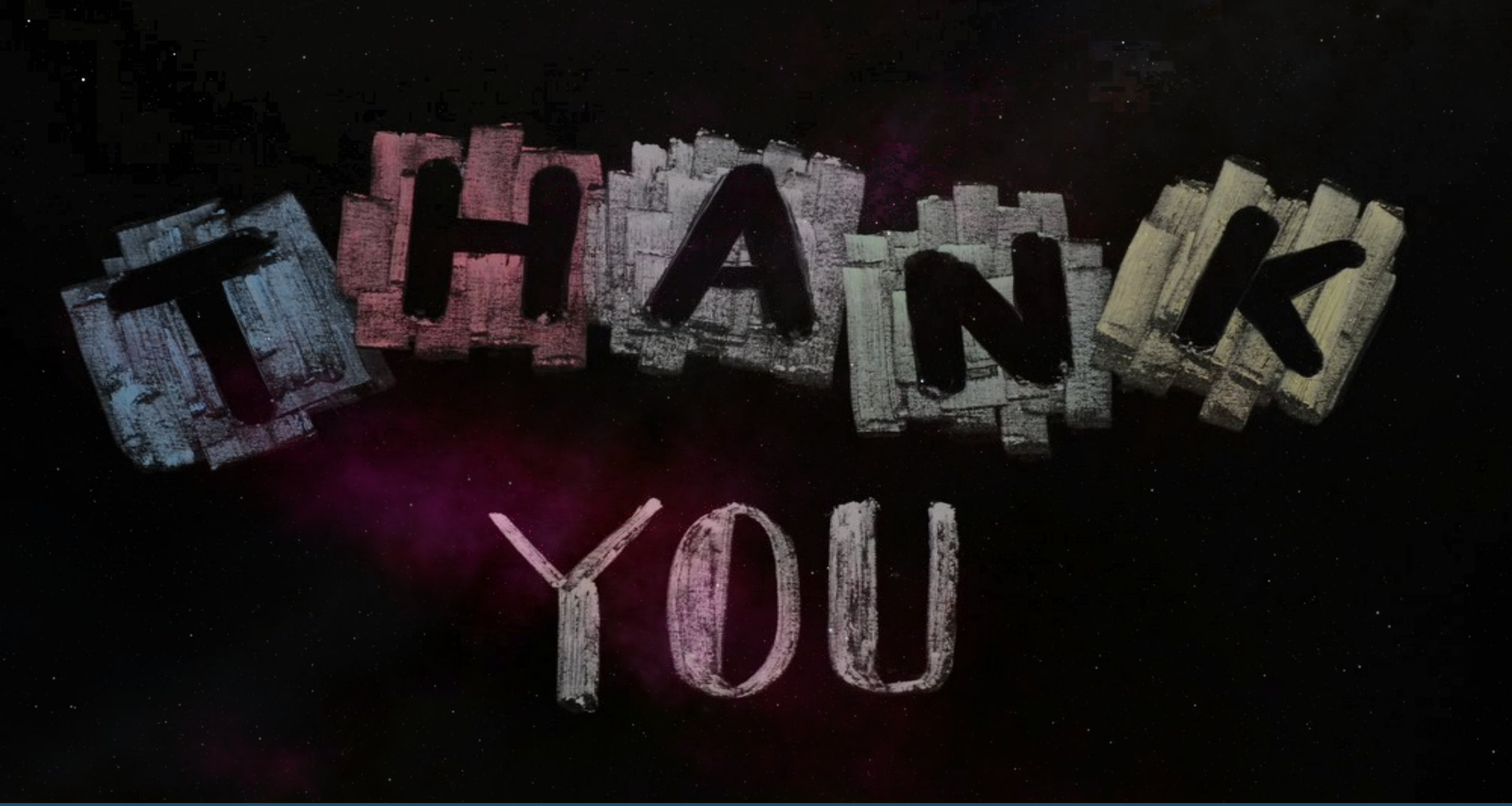
Risk event	Impact/s on project	L	C	R	Risk treatment(s)	Key Responsibility	Project phase
Loss/lack of resources	Project stalls while new resources are acquired and trained. Impacts schedule and budget.	4	5	4	Have each project team member au fait with all aspects of the project	Project technical lead	All phases
Scope creep	Schedule and budget over-runs	5	3	3	Develop a scope management plan with the involvement of Client	Project manager	Planning
Incorrect requirements or specifications	Impacts quality, budget and schedule	4	4	3	Model functional requirements analysis using process reengineering and benchmarking	Project manager, project technical lead	Planning
Inadequate design	Impact on quality	3	3	2	Use software modelling tools	Project technical lead	Design
Incompatible development environment	Impact quality	2	3	2	Research all limitations of development environment and compare with software requirements	Project technical lead	Planning, design, development
Unavailable customer contact	Impact quality	3	2	2	Utilise internet technologies to stay in contact, e.g. e-mail, project web site		All
Problems in coding and unit testing	Impact quality				Have a software inspection process and ensure independent testing is done using strong test cases		Development, testing

Notes: L, likelihood: 5, almost certain; 4, likely; 3, possible; 2, unlikely; 1, rare; C, consequence: 5, severe; 4, major; 3, moderate; 2, minor; 1, negligible; R, risk level: 4, extreme; 3, high; 2, medium; 1, low

Assessment !!! Week Report 3

Reminder:

- Submission Deadline: 17 April 2026, 17:00
- Follow the template on Moodle
- Only team leader need to submit the team report
- The team report link should also shown in your Team GitLab


THANK
YOU